Universe-originating Measurement: Operational, finite-step CCC via $\mathrm{AdS}_4 \leftrightarrow \mathrm{dS}_4$ bridges

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Motivation and idea

Conformal Cyclic Cosmology (CCC) explains the low-entropy start of each aeon by a conformal handover from the previous aeon. This note proposes a *finite-step, operational realization* of that handover: a quantum measurement-like bridge that *creates classical records* and flips an AdS_4 receiver brane into dS_4 aeon brane after finitely many "ticks", *without introducing additional free parameters.*

The starting stance is simple: *Unitarity cannot produce facts.* A purely unitary, invertible channel cannot produce certifiable facts on infinite-time tail. Operationally, the bridge state (AdS side) is *pure*, *factless*, high-energy vacuum.

This pure state AdS_4 brane is built layer by layer from the previous aeon dS_4 brane via pointer-aligned QND (POVM-compatible) channel on the radiative sector and almost-QND gravitational coupling (which produces two-point Diósi-Penrose self-energy scaling) to the canonical commutant KMS bath (hence no spurious heating in dS_4/CFT sector). A Brown–York/anomaly DC rectifier then biases the receiver curvature, driving the AdS_4 brane region to a split that detaches the next dS_4 aeon in a finite number of pulses, "ticks", that have discharged information to the AdS_4 receiver.

(A) Anomaly DC rectifier (trace sector). Linearised about a maximally symmetric background, the DC trace relation reads

$$\gamma\,\langle R^2\rangle \;-\; \langle R\rangle \;+\; 4\Lambda_4 \;=\; 8\pi G_4\,\overline{\tau}\,.$$

For zero-area LoG/LoG ticks the DC component satisfies $\overline{\tau} = 0$, but $\gamma\langle R^2\rangle > 0$ produces a genuine *bias* of the mean curvature. Hence repeated ticks raise $\langle R\rangle$, with exact feedback $\sum \gamma\langle R_k^2\rangle = |R_0|$, making the final pre-split curvature-resolution step arbitrarily small.

Recoverable information increases from aeon to aeon, while the cutoff scale and curvature decrease near-geometrically. In late aeons, the newest KR heads (those at the highest cutoff) decouple; as a UV-accumulating sequence they converge, in the limit, to the true AdS_5 boundary. The approach to this boundary—and the Lorentzian evolution it induces—is self-caused: a CPTP RG flow along the second AdS_5 (radial) time defines a 5D Universe-originating quantum measurement.

(B) Exact exhaustion within an aeon and geometric inheritance across aeons. Let $C_n := |R_0^{(n)}|$ be the initial AdS_4 curvature magnitude inherited by aeon n. Along the solvable one-wall receiver evolution, the Brown–York curvature increments are positive and exactly summable,

$$0 < \Theta_k < |R_k|, \qquad \sum_{k\geq 0} \Theta_k = C_n,$$

so the receiver curvature is resolved monotonically with an arbitrarily small final step. At split, the completion sector carries an inherited share of the exported coherence stock into the newborn dS_4 brane. If that inherited share, measured relative to C_n, remains approximately stable from aeon to aeon, then

$$C_{n+1} \approx q_n\, C_n, \qquad q_n \approx q < 1,$$

and the initial AdS_4 curvature decreases near-geometrically across aeons.

5D Schwinger-Keldysh influence action is minimized, producing the classical worldline across all dS_4 branes as a saddle point, and governing the information transfer structure (*minimized quantum/classical excess* on the source brane, Fisher-quadratic covariances on the receiver, and welded pulse temporal/spatial shapes and locations)

(C) 5D Universe-originating measurement action (on–shell SK & holographic HJ). A single on–shell SK influence for the AdS_5 bulk (EH + GHY + counterterms + GW) with dS source and AdS_4 head reduces to a boundary variational functional for each tick k:

$$\begin{aligned}
\mathcal{A}_k = \quad & \underbrace{F_{\mathrm{dS}}\big[\rho_k \,\big\|\, \rho^{\mathrm{BD}}_{\Delta_k}\big]}_{\text{source: loss of recoverable info}} - \underbrace{C_k[h_k]}_{\text{exported record } (C_k \propto \Delta I_k[h_k])} \\
& + \underbrace{\Phi^{\mathrm{phys}}_{\partial}\big[h_k;\, \Lambda_k\big]}_{\text{receiver: payload packing (incl. GW/portal)}} + \underbrace{\alpha_{\mathrm{W}} \left\|\Pi_{\mathrm{W}}\, \delta C_k\right\|^2_{F,\Lambda_k}}_{\text{Weyl/Riegert (trace/rectifier from EH+ct)}} \\
& + \underbrace{\alpha_{\mathrm{RP}} \left\|\Pi_{\mathrm{RP}}\, \delta C_k\right\|^2_{F,\Lambda_k}}_{\text{OS–odd monitor (vanishes on healthy heads)}} + \underbrace{\alpha_{\mathrm{conf}} \left\|\Pi_{\mathrm{conf}}\, \delta C_k\right\|^2_{F,\Lambda_k}}_{\text{conformal/Ward–residual monitor}} \\
& + \underbrace{\int_{I_k} \frac{1}{2}\, \mu_{\mathrm{RG}}(\Lambda_k)\, \dot{\Lambda}_k^2\, dt}_{\text{cutoff work (radial HJ; EH+ct } \Rightarrow \mu_{\mathrm{RG}}(\Lambda) \sim M_4^2(\Lambda))}
\end{aligned}$$

Here h_k encodes the (LoG/LoG) *payload profile* and $C_k \sim \Delta I_k[h_k]$; $\delta C_k = \mathsf{T}_{\Lambda_k}[h_k]$ is the (Gaussian) receiver covariance tangent; $\Pi_{\mathrm{W}}, \Pi_{\mathrm{RP}}, \Pi_{\mathrm{conf}}$ are the Fisher–orthogonal projectors onto the Weyl/trace (Riegert), OS–odd, and conformal mismatch sectors; $\alpha_\bullet(\Lambda_k)$ are (holographically fixed) spectral weights. The stationarity of $\mathcal{A}_k$ determines both the exported payload h_k (hence C_k) *and* the cutoff history $\Lambda_k(t)$.

(D) Higgs portal: constant mass imprint from the anomaly DC. On the receiver brane the heavy brane boundary fluctuation scalar field $\mathcal{S}$ couples as

$$\mathcal{L} \supset \tfrac{1}{2}(\partial\mathcal{S})^2 - \tfrac{1}{2} m_{\mathcal{S}}^2 \mathcal{S}^2 \;-\; \lambda_p\, \mathcal{S}\, H^\dagger H \;+\; \mathcal{S}\, J_{\mathrm{anom}}\,, \qquad J_{\mathrm{anom}} \;:=\; \kappa\,(\mathcal{K}_{\mathrm{anom}} * R^2)\,.$$

Solving at tree level gives a *constant* post–flip shift

$$\boxed{\Delta\mu_H^2 \;=\; -\frac{\lambda_p}{m_{\mathcal{S}}^2}\, J_{\mathrm{DC}}, \qquad J_{\mathrm{DC}} := \big(\mathcal{K}_{\mathrm{anom}} * R^2\big)_0\,,}$$

so that $v^2 = \big(\lambda_p J_{\mathrm{DC}}/m_{\mathcal{S}}^2 - \mu_{H,0}^2\big)/\lambda_H$. For $m_{\mathcal{S}} \gg v$ the one–loop threshold $\delta\mu_H^2 \sim (\lambda_p^2/16\pi^2) m_{\mathcal{S}}^2$ is technically natural; collider and fifth–force signatures remain quiet ($\delta\kappa_h \ll 10^{-3}$ from the induced-gravity Robin boundary conditions-suppressed visible overlap).

Two theorems about interaction of Lorentzian and RG times are proven. Exactly-preserved classical information of receiver Weyl anomaly record acts as a ledger with approximately classical points of 5D Universe states at the end of each aeon, connected by aeon-to-aeon RG steps. Square in two times commute on records, yet its two canonical completions differ by one intrinsic bit. That bit is carried by neither time alone; it appears only in their mixing. Physically, time mix in a delayed zig-zag between in-aeon completion sector affine charge *$dS_4 \to AdS_4$/Completion sector* and inter-aeon morph+export *Completion sector $\mapsto$ next $dS_4 \to$ next AdS_4*.

(E) Two-time theorem consequences.

$$\mathbf{2T}_{\mathrm{LS}}: \text{ mixing of Lorentzian and RG time yields an intrinsic one-bit holonomy,}$$

giving theorem-level "it from bit" and an objective before/after distinction carried by neither time alone.

$$\mathbf{2T}_{\mathrm{Main}}: \text{ the same odd bit survives on the UV boundary as a local odd channel,}$$

yielding spinor symmetry and full superconformal symmetry on the boundary.

Observational handles

CMB: A degree-scale *compensated LoG/SMHW* imprint (Cold-Spot-like morphology) with a definite *TE sign* at the core scale.

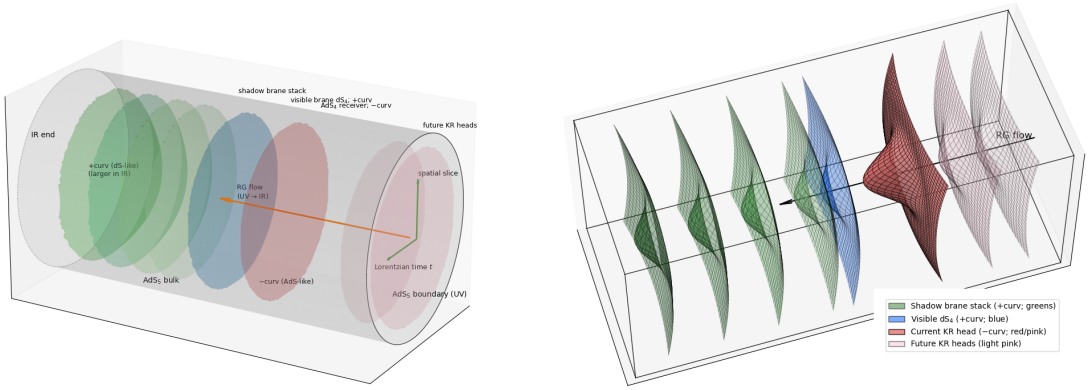

Figure 1: a) AdS_5 bulk with the visible dS_4 brane, receiver AdS_4 brane, stack of shadow dS_4 branes, and future heads (shown in AdS phase). 1D temporal/1D spatial slice and meta-causal RG flow are shown. b) Bulk and spatial slice of branes carrying DC LoG imprints from Brown-York anomaly rectification.

LSS/halos: A *screened low-pass "mirage" matter* contribution

$$\Omega_X(k,a) \sim \Omega_{2,0}\, a^{-3} \times e^{-d(a)\sqrt{k^2+m_\bullet^2}} \times \frac{\sqrt{k^2+m_\bullet^2}}{\sqrt{k^2+m_\bullet^2}+r(\Lambda)k^2}.$$

Core scale $r_{\rm core} \sim \max\{r_{\rm scr},\, m_\bullet^{-1},\, \Lambda^{-1}\}$ matches cluster cores; coupling of cutoff-decreasing shadow brane stack produces log-skew distribution of kpc dwarf cores. Fifth-force bounds are satisfied via curvature-dependent, BC-suppressed visible overlap.

Falsifiability & data-aware wedge

- **CMB (Planck):** LoG/SMHW core with *TE phase*; ring optional/suppressed.
- **CMB+BAO:** early "mirage" fraction $\lesssim$ few % at $z \sim 1100$, decaying to $\lesssim$1% today.
- **Full-shape $P(k)$ (BOSS/DESI):** roll-off begins at low k; coherent modulation at few-% at most.
- **Halos:** dwarfs 0.5–3 kpc, log-skew distributed; clusters $\ll$ 50 kpc.

Links & contact

Theory archive (manuscripts, notes, releases): https://github.com/dmytrosurdu/2T-releases

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